Category:

Crypto

Name:

Attack the bad-box

Message:

You are given some examples of encrypted integers and decoded messages.
Then, please decrypt the integer array below to get the flag.

-48, 35, -55, -93, 63, -103, 81, 85, 54, 1, -108, -4, -16, 77, 5, 58, -5, -57, 15, -106, -112, -92, -119, -126, -55, -80, -75, -27, 41, 27, 11, -71, -59, 33, 112, -75, 19, -95, -1, 50, -127, 18, -121, -21, 103, -87, 11, 24, 103, 84, -16, -53, -90, 127, 60, 79, -97, -127, 54, -64, -56, -106, -72, -22, -88, -126, -29, -125, 125, 92, 93, -51, -89, 19, 69, -35, 73, -26, -87, 11

As always, the flag starts with “CSG_FLAG”.

Instructions:

The encryption is done with XOR and base64.
The assumed steps to solve this problem is the below.

1. You need to check the relationship between encrypted integers and decoded messages. You can notice the base64-encoded decoded message is the half length of the encrypted array.
2. By XORing the string from the head, you see the same string with the second half.
3. So you can see the decryption can be done with (1) XOR the first half with the second half. (2) decode base64.

Another solution

1. Use some chars from the base64-encoded “CSG_FLAG”: Q1NHX
2. XORing the string with the encrypted binary, you can notice that the same binary sequence (as before XOR) appears from position 40: 0xd0, 0x23, 0xc9, 0xa3 and 0x3f.
3. So you can see, after the position 40 is the xor key.

You will get the base64 string: “Q1NHX0ZMQUd7V29udF9VX21ha2VfTGVtb25hZGV9”.
Decode this to get the flag “CSG_FLAG{Wont_U_make_Lemonade}”

Solution script:

```
c = [-48, 35, -55, -93, 63, -103, 81, 85, 54, 1, -108, -4, -16, 77, 5, 58, -5, -57, 15, -106, -112, -92, -119, -126, -55, -80, -75, -27, 41, 27, 11, -71, -59, 33, 112, -75, 19, -95, -1, 50, -127, 18, -121, -21, 103, -87, 11, 24, 103, 84, -16, -53, -90, 127, 60, 79, -97, -127, 54, -64, -56, -106, -72, -22, -88, -126, -29, -125, 125, 92, 93, -51, -89, 19, 69, -35, 73, -26, -87, 11]
print(base64.b64decode(''.join([chr(c[i] ^ c[len(c)//2+i]) for i in range(len(c)//2)]).encode()))
```
